

STROKE

Assessment of upper-limb motor recovery after stroke using a wrist-worn accelerometer digital biomarker

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Existing clinical assessments for upper-limb motor rehabilitation poststroke pose limitations as end points for efficient clinical trials. This study aimed to develop a digital metric for assessing motor recovery using accelerometer data collected in naturalistic environments. We constructed the digital arm performance scale (DAPS) by analyzing ~23,000 hours of data from 215 participants, including healthy individuals and subacute and chronic stroke survivors. We decomposed continuous upper-limb accelerometer data into lower-level movement segments, from which key features were extracted and aggregated using a linear mixed-effects model to produce an interpretable digital biomarker. DAPS demonstrated excellent reliability, sensitivity, concurrent validity, known-groups validity, discriminant validity, and responsiveness. Power analysis indicated that DAPS could reduce the required sample size for clinical trials with upper-limb motor recovery end points by more than 60% compared with traditional assessments. These findings highlight the potential of DAPS as a low-burden, scalable assessment tool for upper-limb motor recovery, with potential applications in both clinical trials and practice.

INTRODUCTION

The availability of valid, reliable, and responsive outcome measures is imperative to support the efficient and effective development of new therapeutic interventions for upper-limb motor rehabilitation after stroke (1, 2). Currently, clinician-performed assessment tools, such as the Fugl-Meyer assessment of the upper extremity (FMA-UE) and action research arm test (ARAT), which evaluate patients' behavioral phenotypes within laboratory environments, are considered the gold standard for domain-specific end points (2). However, they rely on ordinal scales with limited granularity and sensitivity (3), are affected by pronounced ceiling and floor effects (4), and are vulnerable to variability caused by external factors, such as patient motivation or assessors' encouragement during motor tasks (5). Furthermore, they demand substantial resources, including trained specialists for administration (6) and ongoing training to maintain reliability (7), which offers only episodic snapshots of a patient's response trajectory. Critically, these tools fail to capture biologically meaningful phenotypes in ecologically valid settings, despite the time- and environment-dependent nature of neural recovery (8). These shortcomings have also posed barriers to translating research findings into clinical practice, despite numerous completed trials (1, 2).

Digital biomarkers, defined as characteristics measured by digital technologies that indicate normal biological processes, disease-related processes, or responses to exposures or interventions, including therapeutic treatments (9), have emerged as a promising alternative (10, 11). In research on upper-limb function after stroke, wearable device-based outcome measures built on continuous inertial data during

patients' performance of everyday activities could support low-cost, objective, and longitudinal assessment of the motor recovery process (3, 6). These devices typically include integrated accelerometers or gyroscopes to track upper-limb movement. However, because their inception in the early 2000s (12), these approaches have predominantly focused on a unidimensional aspect of motor performance, limb activity levels, which quantify wrist acceleration magnitudes of the stroke-affected limb or compare them with the contralateral limb (13). Although simple, easy to interpret, and exhibiting solid convergent validity, these methods have demonstrated weaker concurrent validity and responsiveness compared with standardized domain-specific end points, likely because of their inability to capture the multidimensional complexity of upper-limb motor behaviors. Consequently, clinician-performed assessments remain the primary choice for end points in clinical trials (2).

To bridge these gaps, we developed a new composite digital biomarker for assessing stroke upper-limb motor recovery, called the digital arm performance scale (DAPS), constructed using accelerometer data from the stroke-affected wrist and a set of machine learning algorithms. Extracting information about biologically meaningful phenotypes from unstructured and naturalistic upper-limb movements poses major challenges because of the wide variation in the types of activity of daily living (ADL) and the diverse ways in which these ADLs are executed, both across and within individuals (13). To that end, we used a theory-driven data analytic method, motivated by theories of motor control and behavior (14–17), which decomposes patients' performed linear movements into a lower-level unit of motor behavior called movement segments (18, 19). By focusing the analysis on these isolated segments of linear movements, the proposed technique could capture meaningful phenotypes, irrespective of the amount or type of ADLs undertaken. Our central hypothesis is that kinematic and temporal characteristics of these movement segments are indicative of upper-limb motor recovery poststroke and can be used to construct a valid, reliable, and responsive digital biomarker.

DAPS, according to the US Food and Drug Administration (FDA)'s BEST biomarker category (20), is a monitoring digital biomarker that can be repeatedly used to assess upper-limb motor recovery in stroke survivors. We envision that DAPS could be generated on a daily basis to track recovery trajectories over time, thereby offering

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the potential to serve as a cost-effective end point in clinical trials. Beyond this primary application, it also holds potential for use in clinical care to bolster more personalized therapy (2, 3) and patients' self-managed rehabilitation (21).

RESULTS

DAPS development and validation using ~23,000 hours of continuous accelerometer data

We used a data-driven approach (fig. S1) to construct DAPS, leveraging continuous inertial data collected by a three-axis accelerometer placed on the stroke-affected wrist. DAPS was developed and validated using ~23,000 hours of accelerometer data from 215 individuals in their naturalistic environments (22). The dataset included three different subpopulations captured across three independent studies (Table 1): 61 subacute-stage stroke survivors (23); 79 chronic-stage stroke survivors (24, 25); and 75 neurologically intact, community-dwelling adults (26, 27).

Subacute and chronic stroke survivors participated in multiple assessments. Subacute participants, receiving conventional rehabilitation therapy, were assessed up to eight times over 24 weeks, whereas chronic participants, enrolled in an 8-week upper-limb rehabilitation clinical trial, completed up to 14 assessments over 20 weeks. Each assessment began with clinical evaluations. Subacute participants completed ARAT and FMA-UE, whereas chronic participants completed only ARAT. Continuous accelerometer data from the stroke-affected limb during the following 24 hours of naturalistic activities were collected and analyzed. Healthy individuals underwent a single cross-sectional session, with accelerometer data from both wrists included in the analysis. Healthy individuals were assigned the highest possible clinical scores without administering the assessments, indicating full motor capacity and no impairment.

A machine learning pipeline integrates continuous movement features into DAPS

The data analytic pipeline consisted of acceleration data preprocessing, movement segment decomposition, and machine learning model training, as previously described (Fig. 1) (18). Briefly, a band-pass filter with cutoff frequencies of 0.25 to 2.5 Hz was applied to attenuate gravitational effects, motion artifacts, and sensor noise (23, 25, 26). A heuristic algorithm then identified linear movements, indicating a relatively straight path within three-dimensional (3D) space (28).

Representative examples of these trajectories are illustrated for both short-duration (fig. S2A) and long-duration (fig. S2B) movements. Linear movements were decomposed into movement segments based on velocity zero-crossings within the two primary axes of motion and clustered into three groups as described in a prior work (19). The groups included the homogeneous set, with smooth, bell-shaped velocity profiles typical of neurologically intact individuals (29); the outlier set, featuring irregular morphologies indicative of impairment-specific motor characteristics (17); and the entire set, encompassing both for comprehensive motor characterization. We visualized the average shapes and representative examples of movement segments from the homogeneous set (fig. S3, A and B) and those from the outlier set (fig. S3, C and D). Kinematic, morphological, and temporal features relevant to stroke-related motor deficits were extracted from each cluster and aggregated at the participant level using statistical functions (19).

The machine learning algorithm was designed to integrate relevant features into a comprehensive indicator of motor status, ensuring that DAPS satisfies core statistical validation criteria of reliability, measurement error, concurrent validity, and responsiveness (30, 31). First, feature pruning was performed to eliminate unreliable and irrelevant features. We then trained a linear mixed-effects regression model with a participant-specific random intercept using the pruned features as fixed effects and the min-max normalized ARAT scores as the dependent variable. In this framework, the fixed effects captured population-level associations between the kinematic features and ARAT scores, supporting concurrent validity. The random intercept accounted for participant-specific variability in baseline ARAT scores, whereas the shared slope modeled the consistent and generalizable relationship between changes in feature values and changes in ARAT scores over time, supporting DAPS's responsiveness. To generate DAPS values for the held-out testing dataset, only the fixed effects were applied. The resulting DAPS values primarily ranged from 0 to 1, with lower values indicating more severe motor impairment, although its theoretical range was unbounded and thus free from floor or ceiling effects.

The model was trained using a combined dataset of subacute and healthy participants, excluding chronic participants. Subacute participants were prioritized for training because they exhibited dynamic longitudinal changes in both motor phenotype and clinical scores due to heightened neuroplasticity, which were essential for developing a responsive digital biomarker. In contrast, chronic

Table 1. Summary of the three datasets used for DAPS's development and validation. Participant demographics, clinical assessment modalities, and accelerometer data volumes across three cohorts. N, number of participants; n, total number of accelerometer recordings. Age is the median [interquartile range (IQR)]; data collection per participant is mean ± SD. Abbreviations: ARAT, action research arm test; FMA-UE, Fugl-Meyer assessment of the upper extremity; acc., accelerometer. Dash (–) indicates assessments not administered.			
	Subacute	Chronic	Healthy
Sample size N (n)	61 (224)	79 (615)	75 (150)
Age; median (IQR)	67 (10)	60 (12.5)	53 (12.5)
Female (male)	26 (35)	27 (52)	40 (35)
Data collection per participant	3.67 ± 2.75 days	7.77 ± 3.64 days	1 ± 0 days
Amount of acc. data	5376 hours	13,860 hours	3600 hours
Clinical assessments	ARAT, FMA-UE	ARAT	–
Application to algorithm	Train/validation/test	Test	Train/validation/test

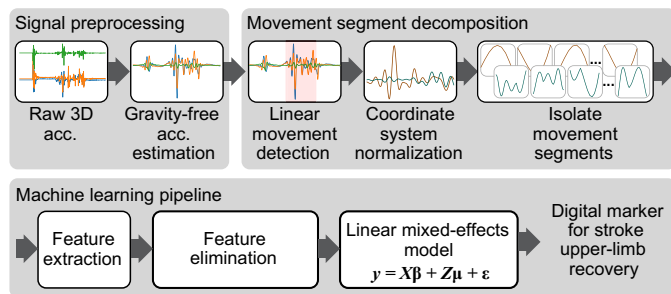


Fig. 1. Development of DAPS from wrist-worn accelerometer data for evaluation of upper arm movement after stroke. Flowchart of DAPS development process where wrist-worn accelerometer (acc.) data were first preprocessed using a band-pass filter to attenuate gravitational components and sensor noise. Linear movements were then identified, and movement segments were decomposed. The machine learning pipeline included feature extraction from movement segments, feature pruning, and training of a linear mixed-effects model.

participants were excluded from training because they demonstrated minimal longitudinal change in ARAT scores, offering limited utility for modeling responsiveness. Healthy participants were included to represent neurologically intact phenotypes. To evaluate performance on the subacute and healthy participants, we used nested leave-one-subject-out cross-validation (LOSOCV), ensuring that data from the same individual were never used across training, hyperparameter tuning, or testing phases (30). Last, to demonstrate the generalizability of DAPS beyond the subacute stage, we used the chronic participants as an independent test set. For this analysis, the model was trained with the subacute and healthy participants by single-layer LOSOCV and applied to the chronic participants.

DAPS demonstrates reliability, validity, and responsiveness in survivors of subacute stroke

DAPS demonstrated excellent within-day split-half reliability in subacute stroke survivors (32), achieving an intraclass correlation coefficient [ICC(2,1)] of 0.95 (Fig. 2A). To assess this, we divided each participant's 24-hour accelerometer data into two groups, alternating between even- and odd-numbered 30-min time windows, and independently computing DAPS from each group. Given that our study yielded only a single 24-hour recording per clinical visit, true day-to-day reliability could not be directly assessed, necessitating this within-day approximation. We also estimated two key metrics for measurement errors—standard error of measurement (SEM) and minimal detectable change (MDC) (33)—and compared them with clinician-performed end points (Table 2). The SEM and MDC of the DAPS score were reduced by 52 to 58% compared with the min-max-normalized ARAT and FMA-UE, indicating greater sensitivity to change.

The DAPS score demonstrated strong concurrent validity against standardized assessments (Fig. 2B). The Spearman's rank correlation coefficients (ρ) were 0.79 with ARAT [$P = 1.6 \times 10^{-82}$, 95% confidence interval (CI) = [0.75, 0.83]] and 0.81 with FMA-UE ($P = 3.8 \times 10^{-89}$, 95% CI = [0.77, 0.84]), indicating excellent agreements (34). DAPS avoids ceiling or floor effects due to its theoretically infinite score range, unlike ARAT and FMA-UE (4).

The DAPS score demonstrated strong known-group validity (Fig. 2C), effectively distinguishing healthy behavioral phenotypes from those of subacute participants ($P = 1.8 \times 10^{-15}$; Wilcoxon rank

sum test), as hypothesized. DAPS values ranged from 0.31 to 1.22 in subacute participants and 0.67 to 1.46 in healthy participants (fig. S4). Furthermore, DAPS was able to distinguish subacute participants who achieved maximum clinical assessment scores from participants with the healthy phenotype, with statistically significant differences ($P = 5.0 \times 10^{-7}$ for ARAT; $P = 4.2 \times 10^{-9}$ for FMA-UE; Wilcoxon rank sum test) (fig. S5, A and B). These findings highlight the ability of DAPS to capture subtle differences in motor phenotypes that conventional clinical assessments cannot resolve due to inherent ceiling effects.

We also assessed the discriminant validity of DAPS regarding age, gender, and whether the dominant or nondominant limb was affected, hypothesizing that DAPS would remain largely independent of these factors. DAPS showed a weak negative correlation with age in both healthy and subacute participants (fig. S6). This finding aligns with the established literature; aging naturally compromises upper-limb performance (35), particularly after age 60, through reductions in strength (36), coordination (37), and movement speed and smoothness (38), in a manner that partially mimics poststroke deficits (39). Therefore, the minor age-related effects observed in DAPS likely reflect genuine physiological declines rather than biomarker bias. The variability in DAPS was predominantly driven by stroke severity, evidenced by strong correlations with ARAT and FMA-UE, rather than age. No significant differences ($P = 2.8 \times 10^{-1}$ and $P = 8.0 \times 10^{-2}$) were observed between genders in either healthy or subacute groups (fig. S7, A and B) nor between affected dominant and nondominant limbs among subacute participants ($P = 2.2 \times 10^{-1}$) (fig. S8A).

The responsiveness of DAPS was examined using three different approaches: correlation between the temporal trajectories of DAPS and standardized assessments, correlations between changes in DAPS scores and changes in standardized assessments, and dichotomous detection of improvement defined on the basis of standardized assessments. DAPS exhibits a strong agreement to standardized assessments for population-level trajectory over the 24-week study period (Fig. 2D). Data were grouped by collection time (defined as weeks from baseline), and the mean trajectories with two SDs were visualized for all assessments. The mean trajectory of DAPS exhibited a Spearman's correlation of 0.98 with ARAT ($P = 3.3 \times 10^{-5}$, 95% CI = [0.87, 1.0]) and 0.93 with FMA-UE ($P = 8.6 \times 10^{-4}$, 95% CI = [0.65, 0.99]). Changes in DAPS between the first and last available visits exhibited Spearman correlation coefficients of 0.56 with ARAT changes (95% CI = [0.29, 0.75], $P = 3.4 \times 10^{-4}$) and 0.41 with FMA-UE changes (95% CI = [0.1, 0.65], $P = 1.2 \times 10^{-2}$) (Fig. 2E), demonstrating moderate-to-strong longitudinal agreement (40).

DAPS also demonstrated capability to detect motor improvements on the basis of external criteria (Fig. 2F). Participants were labeled as responders if the difference in their clinical scores between the first and last visits exceeded the MDC (Table 2) and as nonresponders otherwise. Of the 61 participants, 36 completed more than one assessment, with 14 labeled as responders based on ARAT and 17 based on FMA-UE. We similarly calculated the change in DAPS scores between the same visits, which was compared to a decision threshold to classify participants as responders and nonresponders. By systematically sweeping the decision threshold across the range of DAPS score changes and comparing the results to the labels generated using clinical assessments, we generated the receiver operating characteristic (ROC) curves (Fig. 2F). A leave-two-subjects-out cross-validation strategy was used to estimate variability in ROC performance. DAPS achieved an area under the ROC curve (AUC-ROC) of 0.89 ± 0.01 for

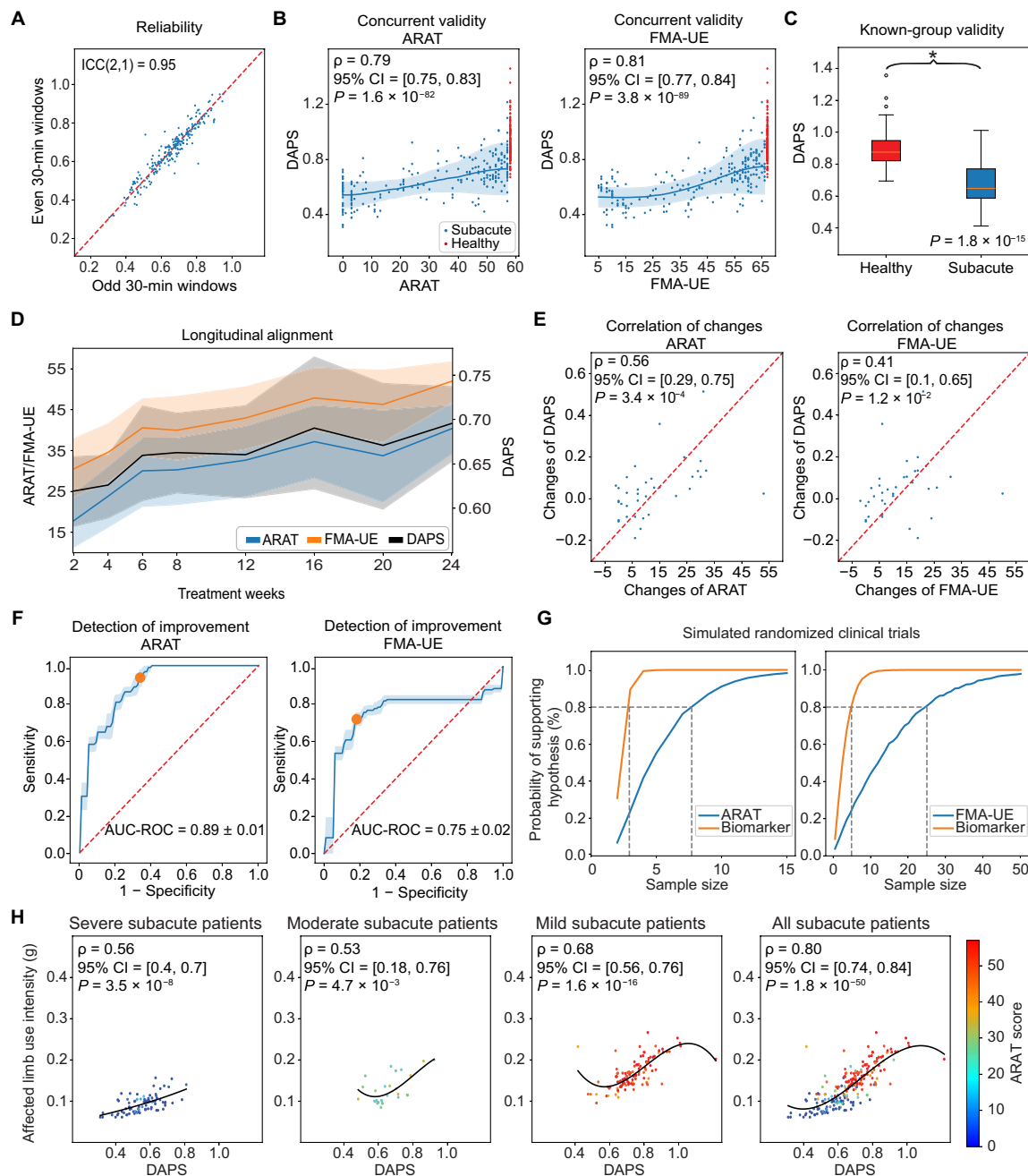


Fig. 2. Evaluation of DAPS in subacute stroke survivors. (A) Scatterplot assessing reliability between even-numbered (y axis) and odd-numbered (x axis) time windows across all subacute stroke survivors' clinical visits. $N = 61$, $n = 224$. (B) Scatterplots showing the Spearman's correlation between DAPS (y axis) and standardized clinical assessments (x axis), including the mean trend (solid line) and two SDs (shaded area). Healthy, $N = 75$, $n = 150$; subacute stroke, $N = 61$, $n = 224$. (C) Box plot comparing the distribution of DAPS between subacute-stage survivors of stroke and healthy control individuals, with statistical significance assessed using the Wilcoxon rank sum test. Healthy, $N = 75$, $n = 150$; subacute stroke, $N = 61$, $n = 224$. (D) Temporal trends of DAPS (black) and ARAT (blue) or FMA-UE (orange) clinical measures over the treatment weeks, showing the mean trend (solid line) and two SDs (shaded region). ARAT, $N = 61$, $n = 224$; FMA-UE, $N = 61$, $n = 224$; DAPS, $N = 61$, $n = 224$. (E) Scatterplots illustrating Spearman's correlations between changes in DAPS (y axis) and changes in clinical assessments (x axis). ARAT, $N = 36$, $n = 36$; FMA-UE, $N = 36$, $n = 36$. (F) ROC curves demonstrating DAPS's ability to detect meaningful improvement in clinical assessments, with mean and one SD obtained by leave-two-subjects-out bootstrapping. The orange dots represent the operation points that maximize F1 scores for further analysis. ARAT, $N = 36$, $n = 36$; FMA-UE, $N = 36$, $n = 36$. (G) Graphical representation of a simulated clinical trial ($N = 61$, $n = 224$) showing the relationship between the probability of supporting the hypothesis (y axis) and required sample size (x axis). (H) Scatterplots depicting the Spearman's correlation between DAPS (x axis) and affected limb use intensity (y axis) across severity groups for subacute stroke survivors. The black solid lines represent the fitted trends of the relationship between use intensity and DAPS, modeled using a cubic polynomial function. Severe, $N = 20$, $n = 82$; moderate, $N = 19$, $n = 27$; mild, $N = 36$, $n = 115$; all subacute, $N = 61$, $n = 224$. Because some stroke survivors transitioned between severity categories across multiple longitudinal visits, the sum of stroke survivors in these subgroups exceeds the total number of unique subacute survivors. Colors indicate functional capacity measured by ARAT, with warmer colors representing greater capacity. N , number of participants; n , total number of accelerometer recordings.

Table 2. Comparison of SEMs and MDCs for DAPS, ARAT, and FMA-UE.

SEMs and MDCs are summarized for the DAPS, ARAT, and FMA-UE in subacute ($N = 61$, $n = 224$) and chronic ($N = 79$, $n = 615$) stroke datasets. To facilitate direct comparison with DAPS, the ARAT and FMA-UE scores were normalized to a 0-to-1 scale by dividing them by their respective maximum possible scores. The SEM and MDC values for the clinical assessments were computed directly from the study cohorts. Dashes (–) indicate uncollected data.

Metric	Measurement	Subacute	Chronic
SEM	DAPS	0.031	0.041
	Normalized ARAT	0.074	0.047
	Normalized FMA-UE	0.064	–
	Actual ARAT	4.25	2.42
	Actual FMA-UE	3.91	–
MDC	DAPS	0.085	0.11
	Normalized ARAT	0.21	0.13
	Normalized FMA-UE	0.18	–
	Actual ARAT	11.77	6.70
	Actual FMA-UE	10.83	–

ARAT and 0.75 ± 0.02 for FMA-UE, both indicative of strong detection performance (40). We further analyzed the misclassified samples at an operating point on the ROC that maximized the F1 scores (indicated by dots in Fig. 2F). Among false responders, five of seven ARAT cases and all three FMA-UE cases showed floor or ceiling effects, specifically, first visit scores of zero or last visit scores near the scale maximum. Among false nonresponders, two of five FMA-UE cases had score improvements comparable to their MDC, suggesting that these may have been borderline or mislabeled cases, whereas the single false nonresponder in ARAT did not exhibit this pattern. These findings highlight the potential of DAPS to mitigate the limitations of traditional clinical assessments in determining patient responses.

DAPS reduces required sample sizes in simulated clinical trials

On the basis of the statistical properties of DAPS, we evaluated potential advantages over traditional clinician-performed end points (ARAT and FMA-UE) in two simulated randomized clinical trials. These simulations followed the Monte Carlo framework proposed by Mori *et al.* (11), which allows for the estimation of trial power under different outcome assessments. The first simulation was designed to examine the efficacy of robotic-assisted therapy versus conventional therapy in chronic stroke survivors, drawing on the trial by Brokaw *et al.* (41), which used ARAT as the primary end point. The second simulation was designed to examine the long-term effectiveness of constraint-induced movement therapy versus conventional therapy in subacute stroke survivors, based on the randomized trial by Boake *et al.* (42), using FMA-UE as the primary end point. Both simulations demonstrated that the choice of outcome measure considerably affects the required sample size to achieve a desired level of statistical power (Fig. 2G). In the first simulation, achieving an 80% probability of supporting the treatment hypothesis required eight participants per arm using ARAT but only three with DAPS, marking a 62.5% reduction. In the second simulation, the required sample size dropped from 26 to 7 participants per arm demonstrating

a 73.1% reduction when using DAPS instead of FMA-UE. We attribute this reduction to the substantially lower SEM of DAPS (Table 2). These findings suggest that DAPS has the potential to reduce the required sample size of clinical trials when adopted as a clinical end point.

DAPS captures motor recovery phenotypes beyond mere limb activity

DAPS was compared to the activity intensity of the stroke-affected limb, computed by accumulating the acceleration magnitude throughout the monitoring period (13, 43). Although affected limb intensity showed a strong correlation with DAPS ($\rho = 0.80$, 95% CI = [0.74, 0.84]), as well as with ARAT ($\rho = 0.82$, 95% CI = [0.79, 0.85]) and FMA-UE ($\rho = 0.84$, 95% CI = [0.81, 0.87]), their relationships varied substantially by stroke severity. Specifically, in participants with severe motor impairment, defined by ARAT scores (44), the correlation was minimal, strengthening progressively only among those with moderate to mild impairment (Fig. 2H). This relationship aligns with the threshold hypothesis proposed by Han *et al.* (45), which suggests that stroke survivors generally refrain from using their impaired limb until reaching a certain level of recovery (46). This adds to the findings supporting that DAPS may capture informative recovery phenotypes rather than merely limb activity.

DAPS integrates multidimensional kinematic features to capture complementary motor performance

The key features that constitute DAPS are summarized to demonstrate its content validity (Table 3). Descriptive functions represent statistical properties derived from movement segments, whereas aggregation functions describe how these descriptive features were synthesized at the participant level. Given that different features were selected in the outer iterations of LOSOCV, we grouped the features into three distinct sets based on their interfeature correlations, similar to factor analysis. Within each of these feature sets, the most frequently selected feature was identified as the key feature for subsequent analysis (19).

The features underlying DAPS capture complementary aspects of motor performance, including velocity, jerk, and the temporal organization of consecutive movements, derived from the full set of movement segments in the secondary direction. Features 1 and 2 quantify maximal motor performance, which aligns with DAPS’s strong association with clinician-rated assessments of maximal functional ability (ARAT) and impairment (FMA-UE) (47). As motor severity improves, participants exhibit more vigorous, active movements, reflected in increases in maximal jerk and high-frequency velocity. The high-frequency velocity component, obtained by applying principal components analysis (PCA) to extract the top five basis functions from the resampled velocity profiles (48), captures the most active portions of movement, similar in concept to high jerk values. Feature 3 quantifies participants’ ability to generate large, alternating upper-limb movement patterns. It was calculated by forming a 2D distribution of median velocities from consecutive movement segments (28) and computing its convex-hull area. This area increased with improved motor status, indicating broader, more alternating movement patterns, whereas more impaired participants produced constrained and repetitive motions (18, 19, 46). Although each feature individually demonstrated acceptable reliability, concurrent validity, and responsiveness, the composite biomarker, integrating these multidimensional features, yielded a more balanced and robust performance across all validation criteria.

Table 3. Key features contributing to DAPS. Summary of the descriptions and clinimetric properties of the key features. “Feature type” denotes the source data from which the features were extracted. “Direction” specifies the axis of linear movement associated with each feature. “Descriptive functions” capture statistical properties derived from the time series of individual movement segments, whereas the “aggregation functions” summarize these descriptive features at the participant level. For features using only descriptive or aggregation functions, the inapplicable function was marked as N/A (not available).

No.	Feature type	Direction	Feature set	Descriptive function	Aggregation function	Reliability ICC(2,1)	Measurement error SEM	Concurrent val. ARAT	FMA-UE	Correlation of changes ARAT	FMA-UE	Detection of improvement (AUC-ROC) ARAT	FMA-UE
DAPS													
1	Jerk	Secondary	All	90th percentile	Median	0.95	0.031	0.79	0.81	0.56	0.41	0.89	0.75
2	High-frequency velocity component	Secondary	All	Maximum	N/A	0.91	0.048	0.82	0.84	0.57	0.49	0.86	0.71
3	Median velocity distribution (v_t versus v_{t+1})	Secondary	All	Convex-hull area	Maximum	0.96	0.045	0.77	0.79	0.43	0.30	0.76	0.71

The reliability and validity of DAPS generalize to an independent chronic stroke cohort

DAPS scores for chronic stroke survivors were generated by applying the machine learning model, trained and validated on the combined subacute and healthy datasets, to the chronic dataset. DAPS showed good reliability in the chronic data, with an ICC(2,1) of 0.86 (Fig. 3A). The SEM and MDC of DAPS were 0.041 and 0.11, respectively, both are smaller than those of min-max-normalized ARAT (Table 2).

DAPS exhibited good concurrent validity with ARAT (Fig. 3B) (34), showing a Spearman’s correlation of 0.65 (95% CI = [0.61, 0.69], $P = 2.3 \times 10^{-93}$). The average trend line between DAPS and ARAT for chronic participants aligned closely with that of the subacute population (Fig. 3C), suggesting that DAPS captured a similar construct across chronicity. However, the variance in DAPS values for a given ARAT score was larger for chronic participants, possibly influenced by individual-level factors, such as psychological, sociological, and environmental factors (23).

DAPS demonstrated strong known-group validity between chronic participants and healthy individuals ($P = 1.2 \times 10^{-20}$; Wilcoxon rank sum test) (Fig. 3D). Detailed value distributions for these groups are available in the Supplementary Materials (fig. S4). Consistent with the subacute findings, DAPS further significantly differentiated healthy phenotypes from chronic patients who received maximum ARAT scores ($P = 5.7 \times 10^{-3}$; Wilcoxon rank sum test) (fig. S5C). For discriminant validity, DAPS showed weak and statistically insignificant ($P = 2.4 \times 10^{-1}$) correlations with age and insignificant ($P = 6.7 \times 10^{-2}$) effect related to limb dominance (figs. S6C and S8B). However, female participants showed slightly higher DAPS scores than male participants ($P = 4.3 \times 10^{-2}$; Wilcoxon rank sum test; fig. S6C). Although the Wilcoxon test accommodates unequal group sizes, our sample was highly imbalanced (27 female versus 52 male) in chronic stroke survivors, which reduces precision for the female subgroup; thus, this difference should be interpreted cautiously and confirmed in a more balanced dataset.

DAPS exhibited limited responsiveness in chronic participants (Fig. 3, E to G). Unlike subacute participants, chronic participants showed minimal motor improvement from the intervention because of reduced neuroplasticity (49) (Fig. 3E). Furthermore, among 69 of the 79 chronic participants with multiple clinical visits, the average change in ARAT from baseline to the final treatment week was 4.7 ± 4.4 , which was smaller than the MDC of 6.7 derived from our data. Similarly, the DAPS’s change was 0.01 ± 0.1 , with an MDC of 0.11. These limitations substantially affected the ability to determine and detect meaningful changes in participant conditions (Fig. 3, F and G). The relationship between DAPS and affected limb intensity paralleled the findings in subacute participants, reinforcing the threshold hypothesis (Fig. 3H).

DAPS implicitly penalizes compensatory behaviors

There is an ongoing debate among rehabilitation clinicians and researchers about whether wearable-based motor assessments should differentiate upper-limb movements performed with normative shoulder and elbow biomechanics from those involving compensatory or substitutive movements (13). Although DAPS was not explicitly trained to assess movement quality, it reliably detects changes (or lack thereof) in motor status (Fig. 2, D to F), showing strong alignment with ARAT and FMA-UE scores, both of which penalize compensatory behaviors (50). For example, we compared chronic

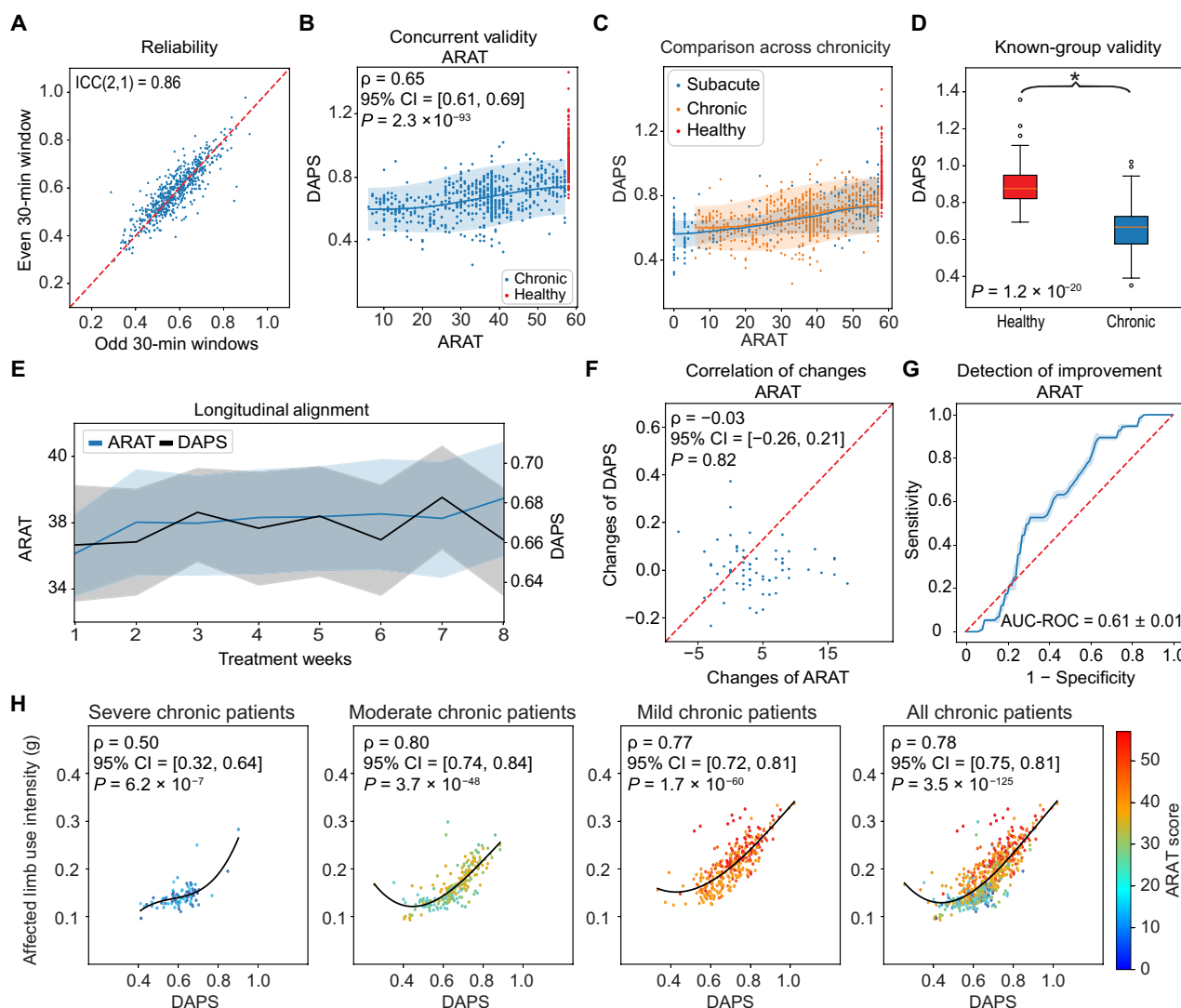


Fig. 3. Validation of DAPS in chronic stroke survivors. (A) Scatterplot investigating reliability between even-numbered (y axis) and odd-numbered (x axis) time windows across all chronic stroke survivors' clinical visits. $N = 79$, $n = 615$. (B) Scatterplot showing Spearman's correlation between DAPS (y axis) and ARAT (x axis). Healthy, $N = 75$, $n = 150$; chronic stroke, $N = 79$, $n = 615$. (C) Scatterplot comparing the distribution of DAPS across subacute (blue), chronic stroke survivors (orange), and healthy participants (red) against their ARAT assessments. Healthy, $N = 75$, $n = 150$; subacute, $N = 61$, $n = 224$; chronic stroke, $N = 79$, $n = 615$. (D) Box plot comparing the distribution of DAPS between chronic-stage stroke survivors and the healthy control group, with statistical significance assessed using the Wilcoxon rank sum test. Healthy, $N = 75$, $n = 150$; chronic stroke, $N = 79$, $n = 615$. (E) Temporal trends of ARAT assessment and DAPS over the treatment weeks, including the mean trend and 2 SDs. ARAT, $N = 79$, $n = 615$; DAPS, $N = 79$, $n = 615$. (F) Scatterplot illustrating the Spearman's correlation between changes in DAPS (y axis) and changes in ARAT (x axis). Chronic stroke, $N = 69$, $n = 69$. (G) ROC curve showing DAPS's ability to detect meaningful improvement in clinical assessments, with mean and one SD obtained by leave-two-subject-out bootstrapping. Chronic stroke, $N = 69$, $n = 69$. (H) Scatterplot depicting the Spearman's correlation between DAPS (x axis) and the affected limb use intensity (y axis) across severity groups and for chronic stroke survivors. Severe, $N = 17$, $n = 93$; moderate, $N = 41$, $n = 217$; mild, $N = 48$, $n = 305$; all chronic, $N = 79$, $n = 615$. Because some stroke survivors transitioned between severity categories across multiple longitudinal visits, the sum of stroke survivors in these subgroups exceeds the total number of unique chronic survivors. Colors indicate functional capacity measured by ARAT, with warmer colors representing greater capacity. N , number of participants; n , total number of accelerometer recordings.

participants' ARAT scores with scores evaluating the degree of compensatory movements during ARAT (fig. S9A), as derived from the study by Barth *et al.* (50), showing that compensatory behaviors were penalized in ARAT scoring. Similarly, DAPS exhibited a moderate, significant negative correlation ($\rho = -0.34$, $P = 2.1 \times 10^{-3}$) with compensatory movement score (fig. S9B), suggesting that individuals exhibiting greater degrees of compensatory behavior in laboratory settings tended to achieve lower DAPS scores. We attribute this to the machine learning model's design, which learned to

identify meaningful phenotypes in motor performance that strongly correlated with clinician-evaluated scores across cross-sectional and longitudinal data.

DISCUSSION

In this study, we developed and validated a digital biomarker, DAPS, for poststroke upper-limb motor recovery by aggregating kinematic characteristics of linear movements during natural behaviors using

a single wrist-worn sensor. DAPS demonstrated excellent reliability, concurrent validity, known-group validity, discriminant validity, and responsiveness and offered clear interpretability, particularly in subacute participants. DAPS also displayed strong generalizability to chronic participants, further supporting its validity across different stages of chronicity.

This research builds on prior in-laboratory studies investigating the kinematics of linear movements during motor recovery, with a specific focus on the theory of submovements (17, 51). These investigations have demonstrated that as impairment severity increases, linear wrist movements performed by stroke survivors on a 2D tabletop become increasingly fragmented and exhibit lower peak velocity (14–17), and accordingly, reduced peak jerk. Conversely, with rehabilitation, movements typically become faster and larger in amplitude, resulting in higher peak velocities and jerk. These kinematic patterns align closely with the features that constitute DAPS (features 1 and 2; Table 3).

Some prior studies have attempted to extend these in-laboratory findings to analyze linear movements during naturalistic upper-limb behaviors (18, 52). However, these studies were conducted in simulated home environments rather than real-world settings and involved only brief periods of data collection (specifically, less than 1 hour per participant). Moreover, they primarily focused on demonstrating concurrent validity without assessing responsiveness or reliability. Despite these methodological differences, our findings corroborate their results, underscoring the importance of variability in movement segment kinematics (feature 3; Table 3) as a critical indicator of motor recovery (18, 19).

Within the International Classification of Functioning, Disability, and Health (ICF) framework (53), DAPS is a measure of motor performance. This construct is distinct from those assessed by standard clinical tools: body function and structure (FMA-UE) and functional capacity (ARAT) (54). Despite these theoretical distinctions, DAPS demonstrates strong cross-sectional and longitudinal associations with these gold standards. This is because DAPS is designed to learn relevant features of motor performance, exhibiting strong cross-sectional and longitudinal associations with clinical measures. Furthermore, unlike existing wearable-based measures of motor performance, which are often limited to unidimensional characteristics of limb activity and may rely on two wrist sensors (13), DAPS provides a comprehensive view of relevant stroke recovery phenotypes only using a single sensor (45, 55).

The findings of this study highlight the substantial capacity of DAPS to streamline clinical trials compared with traditional clinician-performed assessments. Using DAPS as the clinical end point substantially reduces the number of participants required to achieve equivalent statistical power (Fig. 2G). Moreover, DAPS leverages patient-generated data passively collected from a wearable sensor, eliminating the need for frequent clinical visits for motor assessments (55), thereby reducing costs and the burden on both patients and researchers. Collectively, these benefits highlight DAPS's potential to accelerate and optimize clinical trials.

Beyond its utility in streamlining trials, DAPS also offers valuable contributions to clinical practice. The ultimate goal of stroke rehabilitation is to enhance patients' motor performance outside the clinical settings. By evaluating motor behaviors in naturalistic environments, DAPS aligns with this core objective, addressing aspects that traditional in-clinic assessments may overlook. In addition, unlike traditional assessments that offer only sporadic snapshots, DAPS

enables continuous monitoring of upper-limb recovery, a critical feature given the variability in response times to interventions among stroke survivors (8). This capability provides clinicians with timely opportunities to adjust treatment strategies, thereby advancing the vision of truly personalized rehabilitation (2, 3). Moreover, by enabling patients to self-track their own recovery process, DAPS has the potential to enhance their engagement and adherence to prescribed rehabilitation regimens, as supported by prior evidence (21).

This study has several limitations, highlighting important avenues for future research. First, relying solely on three-axis accelerometer data required a high-pass filter to mitigate gravity—a method that impractically assumes constant wrist orientation during movement. Future studies should incorporate gyroscope data to accurately track linear acceleration in a global coordinate system. Second, the healthy control group was slightly younger than the stroke survivors (Table 1) and was assigned maximum clinical scores rather than direct assessment. Although these limitations do not substantially affect the development of DAPS, given that the healthy group primarily served to represent the neurologically intact phenotype, age-related motor differences may have minimally influenced known-groups validity, albeit minimally. Furthermore, the absolute calibration of DAPS may evolve as future datasets incorporate directly assessed, age-matched healthy cohort. Third, the increased variability of DAPS relative to ARAT scores in chronic participants (Fig. 3C) warrants further investigation. Although we posit that individual-level factors may contribute (23, 49), future research should incorporate additional sensors or qualitative assessments to identify the specific drivers of this variance. Fourth, DAPS's responsiveness in the chronic cohort requires further verification. Because chronic participants exhibited minimal longitudinal improvement in ARAT (Fig. 3E), consistent with the biological expectation of reduced neuroplasticity, a definitive responsiveness analysis was precluded. Future studies should prioritize including chronic participants who demonstrate substantial and meaningful longitudinal improvement. Last, several limitations warrant caution regarding the clinical and technical interpretation of DAPS. The analysis of responsiveness in subacute patients (Fig. 2, D to F) is intended to demonstrate methodological validity rather than to define clinical decision thresholds. Similarly, the clinical trial simulations (Fig. 2G) illustrate potential efficiency gains based on data-driven statistical assumptions, rather than reflecting real-world trial outcomes. Furthermore, the reported reliability metrics (ICC, SEM, and MDC) serve as approximations given that they were derived on the basis of the within-day, split-half approach. Regarding technical interpretation, although a consistent set of kinematic features was frequently selected (Table 3), their underlying collinearity requires further investigation. Future work could explore dimensionality-reduced or composite representations to enhance both interpretability and robustness.

In conclusion, we have developed and validated DAPS to assess upper-limb motor recovery in stroke survivors using a single wrist-worn accelerometer, collecting data from patients' naturalistic settings. DAPS demonstrates strong validity, particularly among subacute-phase stroke survivors. These promising results support DAPS's potential as an end point in clinical trials, offering cost effectiveness, low data collection burden, and reduced reliance on specialized clinicians for its large-scale deployment. It also holds promise for clinical practice, where it could help clinicians further personalize treatment strategies (2, 3) and enable patients to self-track and manage their rehabilitation progress (21).

MATERIALS AND METHODS

Study design

This study aimed to develop and validate a digital biomarker (DAPS) for quantifying upper-limb motor impairment in stroke survivors using kinematic features extracted from a single wrist-worn sensor. To achieve this objective, we conducted a secondary, post hoc analysis of data previously collected by our team across three research projects at Washington University in St. Louis, MO, USA (22). The dataset, available through the NICHD DASH repository (56, 57), includes three distinct cohorts (Table 1): (i) an observational study of subacute stroke survivors investigating upper-limb recovery trajectories (23), (ii) a randomized clinical trial (NCT01146379) evaluating the efficacy of task-specific training dosage in chronic stroke survivors (24), and (iii) an observational study of healthy individuals characterizing the relationship between upper-limb use and demographic factors (27).

Across both stroke cohorts, inclusion required a neuroimaging-confirmed first-ever ischemic or hemorrhagic stroke with persistent unilateral upper-limb motor deficits (NIHSS Arm Item scores 1 to 4) (58) and the cognitive capacity to follow multistep commands. Participants were excluded if they had a history of prior stroke, neurological, or psychiatric conditions; severe comorbidities that would confound motor recovery analysis; or pregnancy. Although the subacute cohort captured the full range of early motor deficits, the chronic cohort focused on mild-to-moderate functional capacity (ARAT scores 10 to 48), ensuring participants had at least partial grasp-and-lift capabilities for kinematic validation. Healthy controls were ≥ 30 years old with no history of neurological or physical upper-extremity impairment. Participants were recruited from the Washington University School of Medicine and were stratified into cohorts based on clinical diagnosis rather than randomization. The protocols for the source studies were approved by the Washington University Institutional Review Board, and all participants provided written informed consent before participation (23, 24, 27).

Dataset

For the subacute dataset, data were collected from 61 individuals across up to eight assessment sessions within the first 6 months post-stroke, totaling 224 sessions. These sessions were scheduled at weeks 2, 4, 6, 8, 12, 16, 20, and 24 from baseline. On average, subacute participants completed 3.67 ± 2.75 assessments. At the time of baseline assessment, they were 7.1 ± 3.1 days poststroke. Participants received medical care as per their clinical plan. The chronic dataset consisted of measurement data from 79 stroke survivors at least 6 months post-stroke (on average, 712 ± 1091 days). These patients participated in a clinical trial evaluating the effectiveness of an intervention, completing a total of 615 weekly assessments (7.77 ± 3.64 assessments per participant). Seventy-five percent of these patients received an 8-week intervention, whereas the remaining 25% extended their intervention by up to 4 weeks until reaching a plateau on the primary outcome measure (ARAT). The healthy dataset was obtained from 75 community-dwelling adults without any neurological conditions. Consistent with the prior literature (59, 60), these individuals were assigned maximum scores on the ARAT and FMA-UE scales, without formal administration of the assessments, to represent neurologically intact phenotypes.

For all three groups, ActiGraph GT3X+ accelerometer data were collected from both wrists for ~ 24 hours in their naturalistic environments following clinical assessment. For the subacute and chronic

datasets, only accelerometer data from the stroke-affected limb were included in the analysis, whereas for the healthy dataset, data from both wrists were analyzed. The subacute dataset comprised 5376 hours of accelerometer recordings, the chronic dataset included 13,860 hours, and the healthy dataset contained 3600 hours of recordings. All raw accelerometry data were sampled at 30 Hz. A band-pass filter was applied between 0.25 and 2.5 Hz to approximate gravity-free acceleration and attenuate motion artifacts and inherent sensor noise for subsequent analysis (23, 24).

Linear movement detection and segmentation

Our data analytic pipeline began by identifying linear movements from continuous gravity-free acceleration data. Linear movements refer to upper-limb motions that are predominantly directed along a single axis, such as reaching to grasp an object or transporting it from one location to another. These movements establish the foundation of most ADLs (61) and are widely used in stroke rehabilitation to evaluate motor function because their kinematics effectively reflect motor recovery (16, 17). A heuristic algorithm (fig. S10) was applied to identify linear movements using the following criteria: (i) Signal variance is concentrated in a single eigenvector, indicating a relatively straight path in 3D space, determined by PCA on the 3D gravity-adjusted accelerometer data, where the variance percentage of the first PC exceeds 82% of the total variance. This threshold was derived from manually labeled linear movements in a previous study (18). (ii) Movement duration exceeds 1 s to exclude brief transients and noise, as suggested by Bai *et al.* (62). (iii) The movement magnitude (activity index) exceeds 0.0045g, a threshold recommended by Bai *et al.* (62) to ensure the detection of substantial, meaningful upper-limb movements. Identified linear movements were subsequently projected onto the primary and secondary directions using PCA to capture most signal variance while reducing feature dimensionality and computational complexity. On average, the first PC accounted for $\sim 87\%$ of the variance, whereas the second accounted for about 11% across the three participant groups. As for the amount of linear movement, we observed 291.18 ± 205.05 linear movements per hour in subacute stroke survivors, 365.73 ± 145.72 per hour in chronic stroke survivors, and 612.19 ± 127.95 per hour in healthy participants.

Movement segment decomposition

After identifying linear movements, we derived the velocity time series by integrating acceleration data along both the primary and secondary directions. We then decomposed these velocity time series into finer-grained units, termed movement segments, based on velocity zero-crossings. For each movement segment, we also derived corresponding jerk and displacement segments, defined as the time derivative of acceleration and the time integral of velocity. Research indicates that movement segments contain unique kinematic and temporal patterns relevant to stroke recovery (18, 52), likely resulting from impairments in motor planning and control mechanisms, such as feedforward and feedback control (63, 64). To capture these variations more precisely, movement segments were clustered into three subsets based on the shape of velocity profiles: the homogeneous set, the outlier set, and the entire set. The homogeneous and outlier sets were identified using the unsupervised clustering methods outlined in a prior work (19). This approach builds on findings that the velocity profiles of movement segments in neurologically intact individuals typically exhibit a smooth, bell-shaped curve (29, 65),

whereas stroke survivors often display more irregular morphologies (16, 17). The entire set encompassed all movement segments, combining the homogeneous and outlier sets. We extracted features independently from these subsets to capture nuanced, context-specific characteristics (homogeneous and outlier sets), as well as more comprehensive characteristics (entire set).

Feature extraction

Data features encompassed kinematic, morphological, and temporal characteristics of movement segments in each cluster. Kinematic features were derived from descriptive statistics—such as minimum, maximum, mean, median, SD, 10th percentile, and 90th percentile—calculated on displacement, velocity, acceleration, and jerk profiles of each movement segment. For morphological features, we first resampled the velocity profiles of the segments to a normalized time duration and applied PCA to extract the top five basis functions, representing both high- and low-frequency movement components, as described in our prior study (48). The aforementioned descriptive statistics were then applied to the values of these basis functions to capture the morphological characteristics of movement segments. Temporal features were designed to capture intersegment dynamics (28), so we extracted them solely from the entire set of movement segments. A 2D histogram represented the probability density of consecutive segments, from which geometric features were derived to describe this distribution, including bounding box, convex hull, and eccentricity (66). Descriptive statistics were then applied to the bounding box dimensions and to the convex hull area, perimeter, compactness, and eccentricity. These kinematic, morphological, and temporal descriptive features were processed within each participant to summarize participant-level characteristics using aggregation functions, such as minimum, maximum, mean, median, SD, 10th percentile, and 90th percentile.

Machine learning model

The training process was designed around the three key pillars of biomarker validation criteria: reliability, concurrent validity, and responsiveness. We first pruned unreliable and irrelevant features based on the following criteria: (i) $ICC(2,1) < 0.75$ for reliability (32), (ii) normalized SEM > 0.05 for measurement error (benchmarked against ARAT and FMA-UE), (iii) Spearman's correlation < 0.75 for cross-sectional convergent validity with clinical scores (34), and (iv) Spearman's correlation < 0.35 for the correlation between feature and clinical score changes [representing moderate responsiveness (40, 67)] and AUC-ROC < 0.7 for detecting responders based on the MDC of clinical assessments (40). To streamline training and prevent multicollinearity, interfeature correlations were computed, and lower-performing features within highly correlated pairs (Spearman's correlation > 0.85) were removed.

Next, multivariable linear mixed-effects models with L2 regularization were used to combine the remaining features into a composite biomarker. In our setting, linear mixed-effects models served as an interpretable supervised learning model (68, 69). We modeled pruned kinematic features as fixed effects to capture population-level associations with the normalized ARAT score, which served as the dependent variable. The ARAT score was selected as the primary target variable because it was the only clinical assessment available for chronic participants and strongly correlates with FMA-UE (Spearman's correlation = 0.96 with $P = 8.9 \times 10^{-120}$ from our dataset) (70). A participant-specific random intercept was included to

account for individual differences in baseline motor severity, with each participant treated as a grouping factor. We assumed a shared population-level slope across all participants; that is, although each participant could have a distinct baseline (intercept), the relationship (slope) between kinematic features and ARAT score remained constant. This structure allowed the model to capture both generalizable associations across the population (concurrent validity) and within-participant changes over time (responsiveness). Conceptually, the model followed a two-level hierarchical structure: Level 1 represented the within-participant level, where kinematic features served as fixed-effect predictors, and Level 2 captured between-participant variability via random intercepts. Specifically, only the fixed effects, which represent population-level slopes for kinematic features, were used to generate DAPS score on the unseen testing dataset. It is worth highlighting that we did not intend to use ARAT to devise a digital version of itself as it is not a perfect gold standard for assessing motor recovery. Rather, we used it to capture associations, aligning with best practices in digital biomarker research (47). With min-max normalization applied to the target clinical measure (ARAT), the primary range of DAPS was from 0 to 1, although its theoretical range was unbounded.

The models were trained, validated, and tested using nested LOSOCV, ensuring no data from the same participant was used across these stages. The outer cross-validation was dedicated to testing, whereas the inner cross-validation optimized the L2 regularization parameter. This framework is essential for evaluating the generalizability of digital biomarkers (30).

Evaluation

This study estimated the test-retest reliability [$ICC(2,1)$] of DAPS to assess its consistency against potential day-to-day variability. Because of data constraints of only 24 hours per participant, we split the dataset into two groups by alternating the group categorizing every other 30-min time window (specifically, alternating between even- and odd-numbered time windows). The 30-min window length was chosen on the basis of the American Time Use Survey conducted by the US Bureau of Labor Statistics (71), which indicates that many daily activities occur within 30-min intervals. Partitioning the data into interleaved 30-min subsets minimized overlap of similar activities, sampled a representative range of behaviors across the day, and enabled a more robust estimate of DAPS's test-retest reliability. To quantify measurement error, we estimated the SEM and MDC. SEM represents the inherent error in a measure due to unreliability, whereas MDC estimates the smallest detectable score change that reflects a true difference, rather than random measurement error or variability (33). Detailed equations for these metrics are provided in the Supplementary Materials.

Concurrent validity examines the extent to which an outcome measure (in this case, DAPS) is an adequate reflection of a gold standard (including FMA-UE and ARAT). In this study, we evaluated concurrent validity using Spearman's correlation coefficients. For known-group validity, the Wilcoxon rank sum test was used to compare differences between stroke survivors and healthy participants. Discriminant validity examines if the construct captured by DAPS was different from other constructs such as age, gender, and limb dominance in this study. We computed Spearman's correlation between age and DAPS at baseline and used the Wilcoxon rank sum test to compare DAPS values at baseline between gender groups and between affected dominant and nondominant limbs.

Responsiveness refers to the ability of a measure to detect meaningful changes over time in the construct being assessed. In our study, we validated DAPS using three approaches. First, we evaluated the correlation between the temporal trajectories of DAPS and standardized clinical assessments using Spearman's correlation coefficient. Second, we computed the correlation of changes between DAPS and standard clinical assessments (40) to assess whether DAPS could reflect changes in clinical assessments over the course of therapy. Last, we evaluated whether DAPS could detect improvement, as defined by standardized measures, using AUC-ROC (40). Stroke survivors were labeled as “responders” or “nonresponders” based on whether the change in standardized assessments between their first and last visits exceeded the computed MDC (Table 2). AUC-ROC was then used to examine the agreement between the improvement in DAPS and the classification of patients as responders or nonresponders.

Hypothetical clinical trial

We implemented two hypothetical clinical trials using the statistical simulation framework proposed by Mori *et al.* (11). The first simulation modeled a trial comparing robotic-assisted therapy versus conventional therapy in chronic stroke survivors, mirroring the study by Brokaw *et al.* (41) that used ARAT as the primary end point. The second simulation modeled a comparison between constraint-induced movement therapy and conventional therapy in subacute survivors, based on the trial by Boake *et al.* (42), using FMA-UE as the primary end point.

For each simulation, population means and variances for changes in clinical scores were derived from the respective source studies: ARAT from Brokaw *et al.* (41) and FMA-UE from Boake *et al.* (42). To estimate parameters for DAPS, we assumed DAPS would reflect the same percentage of improvement as the corresponding clinical assessment. However, given that DAPS exhibited substantially smaller SEM compared to the ARAT and FMA-UE, we proportionally reduced the population variance for DAPS. The differences in reliability between DAPS [ICC(2,1) = 0.95] and traditional end points [ICC(2,1) = 0.965 for both ARAT and FMA-UE (70)] were negligible, and therefore its impact on the variance was disregarded for simplicity.

Monte Carlo simulation was performed by sampling means and variances from Gaussian and chi-square distributions, respectively. A two-tailed Welch's *t* test was used to account for unequal variances, and hypothesis support was defined as achieving statistical significance between the intervention and control groups with $\alpha = 0.05$, consistent with the original trials. The simulation involved 10,000 iterations across a range of sample sizes: 2 to 250 per arm for the Brokaw-based scenario and 2 to 60 per arm for the Boake-based scenario. Statistical power (the probability of hypothesis support) was calculated as the proportion of iterations yielding significant differences. This framework estimated the minimum sample size required for each assessment to achieve adequate statistical power.

Supplementary Materials

The PDF file includes:

Methods
Figs. S1 to S10

Other Supplementary Material for this manuscript includes the following:

MDAR Reproducibility Checklist

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they have no competing interests. **Data, code, and materials availability:** All data associated with this study are present in the paper or the Supplementary Materials. All human data used in this study were obtained from the NICHD Data and Specimen Hub (DASH) repository. The deidentified datasets, including raw accelerometry and clinical assessment scores for the subacute, chronic, and healthy cohorts, are available to qualified researchers through the DASH platform at <https://dash.nichd.nih.gov/study/426315> (DOI: 10.57982/fayx-p832) and <https://dash.nichd.nih.gov/study/426433> (DOI: 10.57982/72z7-m179). Access to these data are subject to the standard data use agreement (DUA) required by the NICHD DASH. The trained

DAPS biomarker is publicly available at <https://github.com/Ryan-RE-Wang/DAPS> (Zenodo DOI: 10.5281/zenodo.22287158). No other new materials were used or generated in this study.

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